

Supplementary Material

Causal graph recommendation under exposure bias: removing the repeated-walk bias of doubly robust propagation

Theorem, table, section and equation numbers without the prefix S refer to the main text.

S1. Structural properties of linear propagation

The results of this section use only that, given the nuisances, s is a fixed linear function of the edge estimates. They hold for DRUP and equally for the uncorrected operators and for linear LightGCN, and they describe the recommender with its nuisances held fixed.

Proposition S1 (Affine in the user’s own data). *Fix u and a candidate i with $O_{ui} = 0$, and let $G^{(-u)}$ be G without u ’s own contribution. With frozen nuisances, $s_{ui} = \text{const} + \sum_{j \neq i} \omega_j w_{uj}$ with $\omega_j = b(c_{uj}G_{ji}^{(-u)} + c_{ui}w_{ui}c_{uj}^2)$. Hence un-logging interaction j , i.e., replacing w_{uj} by its unlogged value $\hat{Y}_{uj}$, changes s_{ui} by exactly $\phi_j = \omega_j(w_{uj} - \hat{Y}_{uj})$, removal effects add up over any subset, and completeness holds exactly.*

Because the removal game is additive, the ϕ_j are also its Shapley values; this is a consequence of linearity, and the attribution is relative to the stated baseline (Kumar et al., 2020).

Theorem S2 (Optimal minimal counterfactual explanation). *For candidates i, k with $s_{ui} > s_{uk}$, sorting $g_j = \phi_j^{(i)} - \phi_j^{(k)}$ in decreasing order and removing*

interactions until the cumulative sum exceeds the margin yields a minimum-cardinality set whose removal makes k outrank i . If the positive g_j do not sum above the margin, no such set exists. The cost is $O(n + d_u \log d_u)$.

Theorem S3 (Certificates against injected profiles, frozen nuisances). *With frozen nuisances, one injected user v changes u 's three-hop score of item k by $\Delta_k = x_k(D + a_k c_{vk}) - a_k x_k^2$, where $a = \tilde{w}_u$, $x = c_v \odot w_v$ and $D = \langle a, x \rangle$. If v logs at most L interactions with arbitrary items, labels and propensities $\geq \tau$, then D and x_k lie in closed-form intervals. Δ_k is linear in D and quadratic in x_k , so its extremes $[\underline{\Delta}_k, \overline{\Delta}_k]$ over the box formed by the two intervals are attained at its edges or at the vertex of the quadratic. The box contains every feasible (D, x_k) but ignores that both are determined by the same L interactions, so $[\underline{\Delta}_k, \overline{\Delta}_k]$ contains the feasible range of Δ_k and is conservative rather than exact. Item t provably stays out of u 's top- K against any F such users if $s_{ut} + F\overline{\Delta}_t$ is below the $(K+1)$ -th largest value of $s_{uk} + F\underline{\Delta}_k$ over the other candidates. With unclipped IPS, the influence of a single interaction is unbounded.*

The same argument certifies the uncorrected operators (with $\Delta_k = x_k D$). Degrees taken from $\hat{Y}$ make the fake user's weight c_v known; degrees taken from W would be attacker-controlled.

Theorem S4 (Joint differential privacy with public nuisances). *Let the population-level parts of the nuisances (the propensity model, the item parameters of $\hat{Y}$, the item degrees in C and the radius R), the hyper-parameters and the constants a, b of the score be fixed a priori or computed from public data and from the release, and every user-level part from the user's own row.*

Scale each private row so that $\|\tilde{w}_v\|_2 \leq R$ and $\|c_v^{\odot 2} \odot w_v\|_2 \leq R^2$. Releasing G , computed over the private rows, plus symmetric Gaussian noise of scale $\sigma = \sigma_{\text{aG}}(\epsilon, \delta; 2R^2)$, the analytic Gaussian calibration (Balle and Wang, 2018), is (ϵ, δ) -DP under adding or removing one user, for every $\epsilon > 0$. Scoring by (4) uses only the release and the user's own row, so the recommender is (ϵ, δ) -jointly DP (Hsu et al., 2014). Post-processing such as low-rank denoising is free.

Row scaling changes the estimate for users whose rows exceed R , so Theorems 1 and 7 hold for the private variant only up to this shrinkage.

Theorem S5 (Exposure-capped allocation). *Let N users each receive K of their candidates and let item i be recommended to at most cap_i users. Let $\hat{x}$ maximise $\langle x, s \rangle$ over such allocations. The constraint matrix is totally unimodular, so the linear relaxation has an integral optimum, and min-cost flow on the scores rounded to a grid of L levels finds $\hat{x}$ exactly up to the rounding gap $\Gamma \leq NK(\max s - \min s)/L$. All caps hold, at least $NK/\max_i \text{cap}_i$ items are recommended, and for the full-exposure utility $U(x) = \langle x, F^* \rangle$,*

$$U(\hat{x}) \geq \max_x U(x) - \Gamma - 2 \max_x |\langle x, s - F^* \rangle|.$$

If $\mathbb{E}s = F^*$ (DRUP under Assumption 2 with $\xi \equiv 0$), McDiarmid's inequality and a union bound over all mn pairs give $\max_x |\langle x, s - F^* \rangle| \leq NK t_\vartheta$ with probability $1 - \vartheta$, where

$$t_\vartheta = \max_{u,i} \left(\frac{1}{2} \sum_e \bar{c}_e(u, i)^2 \ln(2mn/\vartheta) \right)^{1/2},$$

$$\bar{c}_e(u, i) = b c_e(u, i) + a C_{ui} \varepsilon_{ui} \mathbb{K}[e = (u, i)]/\tau,$$

$c_e(u, i)$ is the constant of Theorem 6(d) for the three-hop term and the second part is the sensitivity of the one-hop term. The union runs over all pairs because the candidate set depends on the log. For logged-graph propagation, $\mathbb{E}s = F^*(P \odot Y)$, and the last term does not vanish.

S1.1. Proofs

Proposition S1 and Theorem S2. With frozen nuisances, substitute $G = G^{(-u)} + \tilde{w}_u^\top \tilde{w}_u - \text{diag}(\tilde{w}_u^2) + \text{diag}(c_u^2 w_u)$ into (4). The terms $\tilde{w}_{ui} \sum_j \tilde{w}_{uj}^2$ cancel against the local correction, leaving $\tilde{w}_{ui} \sum_j c_{uj}^2 w_{uj}$, and every remaining term involving w_{uj} ($j \neq i$) is linear, with coefficient $c_{uj} G_{ji}^{(-u)} + c_{ui} w_{ui} c_{uj}^2$ in $\hat{T}_{ui}$. Since $O_{ui} = 0$, the one-hop term $a\tilde{W}_{ui}$ does not involve w_{uj} , so the coefficient in $s_{ui} = a\tilde{W}_{ui} + b\hat{T}_{ui}$ is ω_j . Un-logging j replaces w_{uj} by $\hat{Y}_{uj}$, which changes s_{ui} by $\omega_j(w_{uj} - \hat{Y}_{uj})$; the game is additive, so its Shapley values are the individual effects. For Theorem S2, the margin after removing S is $s_{ui} - s_{uk} - \sum_{j \in S} g_j$, so minimising $|S|$ under $\sum_S g_j > \text{margin}$ selects the largest g_j . $\square$

Theorem S3. Adding row v adds contrib_v to G and leaves u 's local terms unchanged, so $\Delta = \tilde{w}_u \text{contrib}_v = (a \cdot x)x - a \odot x^2 + a \odot c_v \odot x$. A logged DR entry satisfies $w \in [\hat{Y} - \hat{Y}/\tau, \hat{Y} + (1 - \hat{Y})/\tau]$ for any label and any $p \geq \tau$, and an unlogged one equals $\hat{Y}$. D is maximised (minimised) by assigning the L slots to the largest positive (negative) deviations, and x_k ranges over the values of a logged or unlogged entry. Every feasible profile gives a point (D, x_k) of the product of the two intervals, so the extremes of Δ_k over this box bound its feasible extremes; they are not attained in general, because the box treats D and x_k as free although both depend on the same at most L entries. For fixed x_k the effect is linear in D , so its extremes over the

box are at the ends of the D interval; for fixed D it is quadratic in x_k , with extremes at the ends or at the clamped vertex. With frozen nuisances G is a sum of per-user contributions, so effects add over fake users. For unclipped IPS, $|W_e| = 1/p_e$ is unbounded. $\square$

Theorem S4.. Because the population-level nuisances are public and every user-level part depends only on the user's own row, adding or removing user v changes G by contrib_v alone. With the row scaling, $\|\text{contrib}_v\|_F \leq \|\tilde{w}_v \tilde{w}_v^\top - \text{diag}(\tilde{w}_v^2)\|_F + \|c_v^2 w_v\|_2 \leq R^2 + R^2$, so the ℓ_2 -sensitivity of the upper triangle is at most $2R^2$. The analytic Gaussian mechanism is (ϵ, δ) -DP for every $\epsilon > 0$ exactly when $\Phi(\frac{\Delta}{2\sigma} - \frac{\epsilon\sigma}{\Delta}) - e^\epsilon \Phi(-\frac{\Delta}{2\sigma} - \frac{\epsilon\sigma}{\Delta}) \leq \delta$ (Theorem 8 of Balle and Wang (2018)); we use the smallest such σ . The classical calibration $\sigma = \Delta \sqrt{2 \ln(1.25/\delta)}/\epsilon$ (Dwork and Roth, 2014) is only valid for $\epsilon < 1$. Each user's output is a function of the release and the user's own data, which gives joint DP by the billboard lemma (Hsu et al., 2014). $\square$

Theorem S5.. The allocation polytope is defined by the incidence matrix of the bipartite user-candidate graph with degree bounds, which is totally unimodular, so the linear relaxation has an integral optimal vertex and min-cost flow finds it. Rounding changes each score by at most $h/2$ with $h = (\max s - \min s)/L$. If x^* is optimal for s and $\hat{x}$ for the rounded scores $\tilde{s}$, then $\langle x^*, s \rangle \leq \langle x^*, \tilde{s} \rangle + NK h/2 \leq \langle \hat{x}, \tilde{s} \rangle + NK h/2 \leq \langle \hat{x}, s \rangle + NK h$, so $\Gamma \leq NK h$. Each item is recommended at most $\max_i \text{cap}_i$ times and NK slots are filled, which gives the coverage bound. With x_F optimal for U , $U(\hat{x}) \geq \langle \hat{x}, s \rangle - |\langle \hat{x}, s - F^* \rangle| \geq \langle x_F, s \rangle - \Gamma - |\langle \hat{x}, s - F^* \rangle| \geq U(x_F) - \Gamma - 2 \max_x |\langle x, s - F^* \rangle|$. For the concentration step, $s_{ui} = a\tilde{W}_{ui} + b\hat{T}_{ui}$ is a function of the independent exposures. Changing O_e moves $b\hat{T}_{ui}$ by at most $b c_e(u, i)$

Table S1: Exact explanations (Proposition S1, Theorem S2) on 200 random test users: largest relative error of completeness and subset additivity against re-propagation; rank of the top-1 item after un-logging the five interactions with the largest attribution, five random ones, or the five most similar ones (0 = still first); share of top-1 recommendations for which no subset of the user’s own interactions makes the second item overtake it; median size of the minimal counterfactual explanation otherwise. Every explanation was verified by re-scoring.

Data	max. rel. error	top-1 rank after 5 removals (attr. / random / similarity)	no CF explanation	median CF size
Coat	6.1e-16	0.04 / 0.01 / 0.01	94%	2.0
Yahoo! R3	8.7e-16	1.65 / 0.38 / 1.40	10%	1.0
KuaiRec	3.1e-06	0.03 / 0.01 / 0.01	52%	7.0

(proof of Theorem 6(d)) and, for $e = (u, i)$, moves $a\tilde{W}_{ui}$ by at most $aC_{ui}\varepsilon_{ui}/\tau$, so s_{ui} has bounded differences $\bar{c}_e(u, i)$. If $\mathbb{E}s = F^*$, McDiarmid’s inequality gives $\Pr(|s_{ui} - F_{ui}^*| \geq t) \leq 2\exp(-2t^2/\sum_e \bar{c}_e(u, i)^2)$. The candidate set depends on O , so we take a union bound over all mn pairs: with probability $1 - \delta$, $|s_{ui} - F_{ui}^*| \leq t_\delta$ for every pair, in particular for every candidate, and an allocation sums NK such terms. $\square$

S2. Structural properties on real data

The properties of Section S1 hold for the model with frozen (explanations, certificates) or public (privacy) nuisances, and for any linear operator; we check them for DRUP and report where they are informative.

Explanations. Attributions recomputed against full re-propagation are exact to floating-point precision (relative error 10^{-16} in float64 on Coat and Yahoo! R3, $3 \cdot 10^{-6}$ in float32 on KuaiRec; Table S1), and removing the interactions with the largest attribution moves the top item further down than

removing random or similar ones, so the attributions are faithful. The counterfactual analysis mostly certifies that no explanation from the user’s own data exists: on Coat 94% of the top recommendations cannot be overturned by removing any subset of the user’s own interactions, on KuaiRec 52%, and on Yahoo!R3 10%; the minimal explanation otherwise has a median size of 2, 7 and 1 interactions. Such recommendations rest on population-level evidence (through G), which the attribution to a user’s own interactions does not explain.

Robustness. No certificate was violated in any run (Table S2). With imputation-based degrees, DR adjacency and DRUP certify every Coat user against one or two fake profiles and 93–95% of users on Coat and KuaiRec against five, and realised attacks with up to 100 fake users placed the target in a KuaiRec top-20 for at most 0.3% of users under IPS, DR adjacency and DRUP (2.7% for the logged graph). Certificates are informative only for small budgets: at $F = 20$ they cover 10–26% of users, and at 200 fake users the target reaches 39% of KuaiRec users under DR and DRUP, more than under IPS (10%) and the logged graph (29%). On Yahoo!R3, where DRUP selects IPS edges with cross-fitted W -degrees, the fake users’ degrees are attacker-controlled and no user is certified. When the nuisances are re-fitted on the poisoned log, which the certificates do not cover, none of the certified user–target pairs of DRUP (5,196 on Coat, 2,821 on KuaiRec) was violated, and the realised hit rates were lower than with frozen nuisances (Table S10 in the additional results section), because the fake users’ own imputation raises their degree and lowers their weight.

Privacy. Table S3 reports the jointly private variant of Theorem S4, in which only public users inform the population-level nuisances and the score constants c_1, c_3 . The configuration is fixed a priori (clip 0.1, ridge 20, additive imputation, $\alpha = 0.5$, control-variate weight 1, $\beta = 1$), not selected on the test data. Useful privacy needs a large user base and a weak guarantee, and the public-nuisance model can itself be the limit. On KuaiRec, with 717 public users, private DRUP reaches nDCG@20 of 0.618 at $\epsilon = 16$ and 0.598 at $\epsilon = 8$, against 0.634 without noise, 0.637 with all nuisances and 0.577 for popularity; it falls below popularity for $\epsilon \leq 4$. On Coat (29 public users) the noise-free model reaches 0.548, only just above popularity (0.527), and it is below popularity at every finite ϵ (0.438 at $\epsilon = 16$). On Yahoo!R3 the noise-free model with public nuisances reaches 0.492 and so is below popularity (0.525) at every ϵ , although the non-private model with all nuisances reaches 0.652: 1,540 public users do not estimate the item parameters of a 2%-dense log. The privacy variant is therefore useful only on the largest log. Privatising the nuisance models with a composed budget is left open.

S2.1. Exposure-capped allocation

Table S4 applies the exposure-capped allocation of Theorem S5 to every operator. The min-cost-flow solver returns the optimum up to its rounding gap (a relative gap below $4 \cdot 10^{-5}$ in every run), all caps hold, and coverage never falls below the guaranteed $NK/\max \text{ cap}$. On KuaiRec, where exposure is extremely concentrated, DR adjacency and DRUP are the most accurate operators under every cap, for example 0.199 against 0.173 (logged graph) and 0.178 (IPS) at $c = 5$, and 0.140 against 0.115 and 0.121 at $c = 2$; the two agree to the third decimal under every cap, so the correction does

not change the allocation either. Under the same cap, however, their lists remain more concentrated (Gini 0.80 against 0.72 at $c = 5$), and at equal Gini the frontier over each operator’s grid (Table S9) favours the correlation-based operators on Coat. On Coat, whose random test items are already diverse, capping costs DRUP more accuracy than it costs the logged graph; on Yahoo!R3 exposure is not concentrated (Gini 0.27) and caps down to twice the fair share cost every operator at most 0.002. The allocation is thus the mechanism that controls concentration, and the DR graph is the better input to it when exposure in the log is very uneven.

S3. Additional results

References

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Table S2: Injection attack: F fake users log a low-popularity target and the most popular items. Realised hit rate of the target in real users’ top- K , and the fraction of real users for whom Theorem S3 certifies that the target cannot enter the top- K against *any* F profiles of that size (frozen nuisances). No certificate was violated.

Data	Operator	realised hit rate			certified fraction		
Coat	$F =$	1	5	20	1	5	20
	logged graph	0.004	0.009	0.444	0.00	0.00	0.00
	IPS adjacency	0.001	0.269	0.661	0.15	0.00	0.00
	DR adjacency	0.000	0.000	0.029	1.00	0.94	0.11
	DRUP	0.000	0.000	0.028	1.00	0.93	0.10
Yahoo! R3	$F =$	1	5	100	1	5	100
	logged graph	0.065	0.065	0.199	0.12	0.00	0.00
	IPS adjacency	0.065	0.065	0.171	0.00	0.00	0.00
	DR adjacency	0.065	0.065	0.171	0.00	0.00	0.00
	DRUP	0.065	0.065	0.171	0.00	0.00	0.00
KuaiRec	$F =$	1	5	200	1	5	200
	logged graph	0.002	0.002	0.288	0.00	0.00	0.00
	IPS adjacency	0.001	0.001	0.105	0.98	0.61	0.00
	DR adjacency	0.000	0.000	0.393	1.00	0.95	0.00
	DRUP	0.000	0.000	0.393	1.00	0.95	0.00

Table S3: nDCG of jointly private DRUP ($\delta = 10^{-5}$, analytic Gaussian mechanism, denoising rank 32 fixed in advance). Population-level nuisances come from a public 10% of the users (non-test users where available; on Coat, whose users are all test users, the public users are excluded from evaluation); “all nuisances” is the non-private model with all nuisances fitted on the whole log. Other ranks in Table S11.

Data	$\epsilon = 0.5$	1	2	4	8	16	∞	all nuisances	popularity
Coat	0.382	0.385	0.403	0.385	0.401	0.438	0.548	0.555	0.527
Yahoo! R3	0.481	0.490	0.491	0.493	0.492	0.493	0.492	0.652	0.525
KuaiRec	0.174	0.175	0.154	0.412	0.598	0.618	0.634	0.637	0.577

Table S4: Exposure-capped re-ranking (Theorem S5, exact min-cost flow). Each item may be recommended to at most c times its fair share NK/n of users. Cells: nDCG@ K (Gini@ K). Every cap held; the full range of caps and the accuracy–concentration frontier are in the additional results section.

Data	Operator	no cap	$c = 5$	$c = 2$
Coat	logged graph	0.502 (0.34)	0.501 (0.34)	0.494 (0.32)
	IPS adjacency	0.507 (0.35)	0.503 (0.34)	0.493 (0.33)
	DR adjacency	0.489 (0.60)	0.489 (0.60)	0.475 (0.48)
	DRUP	0.489 (0.60)	0.489 (0.60)	0.476 (0.48)
Yahoo! R3	logged graph	0.658 (0.28)	0.658 (0.28)	0.658 (0.27)
	IPS adjacency	0.656 (0.27)	0.656 (0.27)	0.656 (0.27)
	DR adjacency	0.656 (0.27)	0.656 (0.27)	0.656 (0.27)
	DRUP	0.656 (0.27)	0.656 (0.27)	0.656 (0.27)
KuaiRec	logged graph	0.603 (0.99)	0.173 (0.72)	0.115 (0.44)
	IPS adjacency	0.612 (0.99)	0.178 (0.79)	0.121 (0.49)
	DR adjacency	0.639 (0.99)	0.199 (0.80)	0.140 (0.50)
	DRUP	0.638 (0.99)	0.199 (0.80)	0.140 (0.50)

Table S5: Stress tests outside Assumption 1 and with misspecified propensities (log-normal noise with $\sigma = 0.5$). Relative $|\text{bias}|$ on unexposed candidates.

Scenario ($K = 3$, rel. $ \text{bias} $ on unexposed candidates)	IPS	IPS + WC	DR	DRUP
independent exposures (Assumption 1)	0.029	0.029	1.137	0.019
fixed-size slates per user (Proposition 4)	0.219	0.219	1.119	0.024
misspecified propensities, exact imputation	0.568	0.568	0.000	0.000
misspecified propensities, noisy imputation	0.568	0.568	1.916	0.158
independent exposures (Assumption 1), $K = 5$	4.667	0.059	5.053	0.040
fixed-size slates per user (Proposition 4), $K = 5$	2.392	0.416	4.633	0.032
misspecified propensities, noisy imputation, $K = 5$	19.266	1.162	16.027	0.204

Table S6: Nuisance protocol in the 40-user, 60-item model of Table 3 with true and with estimated (logistic exposure model, misspecified here) propensities, nuisances from an independent log, cross-fitted, or fitted on the same log without cross-fitting, and degrees from $\hat{Y}$ or from edge estimates. Relative $|\text{bias}|$ of the three-hop term and of the full score with data-dependent constants (3,000 replications).

Propensities	Nuisances	degrees	IPS edges			DR edges		
			uncorr.	DRUP	$s, \text{ data } c$	uncorr.	DRUP	$s, \text{ data } c$
true	independent log	$\hat{Y}$	2.15	0.10	0.05	1.90	0.09	0.35
	independent log	W	5.04	0.25	0.10	3.59	0.18	0.35
	cross-fitted	$\hat{Y}$	0.39	0.64	0.15	0.61	0.80	0.62
	cross-fitted	W	3.12	0.81	0.11	4.11	0.73	0.49
	same log	$\hat{Y}$	0.42	0.70	0.18	0.59	0.70	0.33
	same log	W	0.94	0.97	0.49	0.80	0.86	0.37
estimated	independent log	$\hat{Y}$	29.57	3.70	0.15	25.57	2.11	0.53
	independent log	W	92.66	7.44	0.15	47.74	3.00	0.50
	cross-fitted	$\hat{Y}$	1.29	0.48	0.11	3.41	1.23	0.92
	cross-fitted	W	83.73	0.49	0.18	27.63	0.73	0.59
	same log	$\hat{Y}$	0.60	0.82	0.21	0.64	0.72	0.33
	same log	W	0.93	0.97	0.53	0.83	0.88	0.38

Table S7: Tightness of Theorem 6 in a 12-user, 15-item model (20,000 replications, all propensities $\geq \tau$), over all (u, i) .

	$\tau = 0.3$	$\tau = 0.1$
variance bound (b) / Monte-Carlo variance, median	1582	8422
variance bound (b) / Monte-Carlo variance, minimum	471	2126
McDiarmid half-width (d) / empirical 95% quantile, median	131	1831
McDiarmid half-width (d) / empirical 95% quantile, minimum	73	685

Table S8: Exposure-capped re-ranking for all caps. Cells: nDCG@ K (Gini@ K).

Coat	none	$c = 20$	$c = 10$	$c = 5$	$c = 3$	$c = 2$	$c = 1.5$
logged graph	0.502 (0.34)	0.494 (0.34)	0.501 (0.34)	0.501 (0.34)	0.493 (0.33)	0.494 (0.32)	0.487 (0.28)
IPS adjacency	0.507 (0.35)	0.497 (0.35)	0.506 (0.35)	0.503 (0.34)	0.504 (0.36)	0.493 (0.33)	0.486 (0.29)
DR adjacency	0.489 (0.60)	0.489 (0.60)	0.489 (0.60)	0.489 (0.60)	0.485 (0.58)	0.475 (0.48)	0.444 (0.37)
DRUP	0.489 (0.60)	0.489 (0.60)	0.489 (0.60)	0.489 (0.60)	0.485 (0.58)	0.476 (0.48)	0.443 (0.37)
KuaiRec	none	$c = 20$	$c = 10$	$c = 5$	$c = 3$	$c = 2$	$c = 1.5$
logged graph	0.603 (0.99)	0.300 (0.91)	0.228 (0.84)	0.173 (0.72)	0.139 (0.60)	0.115 (0.44)	0.102 (0.31)
IPS adjacency	0.612 (0.99)	0.307 (0.94)	0.234 (0.89)	0.178 (0.79)	0.146 (0.66)	0.121 (0.49)	0.107 (0.34)
DR adjacency	0.639 (0.99)	0.335 (0.95)	0.257 (0.90)	0.199 (0.80)	0.165 (0.67)	0.140 (0.50)	0.125 (0.35)
DRUP	0.638 (0.99)	0.335 (0.95)	0.256 (0.90)	0.199 (0.80)	0.164 (0.67)	0.140 (0.50)	0.125 (0.35)
Yahoo! R3	none	$c = 20$	$c = 10$	$c = 5$	$c = 3$	$c = 2$	$c = 1.5$
logged graph	0.658 (0.28)	0.659 (0.28)	0.660 (0.28)	0.658 (0.28)	0.656 (0.28)	0.658 (0.27)	0.652 (0.23)
IPS adjacency	0.656 (0.27)	0.657 (0.27)	0.657 (0.27)	0.656 (0.27)	0.656 (0.27)	0.656 (0.27)	0.651 (0.22)
DR adjacency	0.656 (0.27)	0.657 (0.27)	0.657 (0.27)	0.656 (0.27)	0.656 (0.27)	0.656 (0.27)	0.651 (0.22)
DRUP	0.656 (0.27)	0.657 (0.27)	0.657 (0.27)	0.656 (0.27)	0.656 (0.27)	0.656 (0.27)	0.651 (0.22)

Table S9: Best nDCG over each operator’s hyper-parameter grid under a cap on the Gini coefficient of top- K exposure.

Data	Operator	Gini ≤ 0.3	≤ 0.4	≤ 0.5	≤ 0.6	no cap
Coat	logged graph	–	0.492	0.492	0.492	0.492
	IPS adjacency	–	0.495	0.495	0.495	0.495
	DR adjacency	–	0.417	0.462	0.486	0.498
	DRUP	–	0.417	0.441	0.484	0.497
Yahoo! R3	logged graph	0.659	0.659	0.659	0.659	0.659
	IPS adjacency	0.657	0.657	0.657	0.657	0.657
	DR adjacency	0.657	0.657	0.657	0.657	0.657
	DRUP	0.657	0.657	0.657	0.657	0.657
KuaiRec	logged graph	–	–	–	–	0.603
	IPS adjacency	–	–	–	–	0.612
	DR adjacency	–	–	–	–	0.633
	DRUP	–	–	–	–	0.633

Table S10: Certificates under re-fitted nuisances: the nuisances are re-fitted on the poisoned log (no cross-fitting) and the realised top- K is checked against the frozen-nuisance certificates of the clean system.

Data	Operator	fake users F	hit rate (frozen / re-fitted)	certified pairs	violated
Coat	IPS	1, 2, 5, 10	0.001/0.001, 0.018/0.001, 0.268/0.001, 0.536/0.001	217	0
Coat	DR	1, 2, 5, 10	0.000/0.000, 0.000/0.000, 0.000/0.000, 0.004/0.000	5193	0
Coat	DRUP	1, 2, 5, 10	0.000/0.000, 0.000/0.000, 0.000/0.000, 0.004/0.000	5196	0
Yahoo! R3	IPS	1, 5, 20, 100	0.065/0.065, 0.065/0.065, 0.066/0.064, 0.174/0.064	0	0
Yahoo! R3	DR	1, 5, 20, 100	0.065/0.065, 0.065/0.065, 0.066/0.064, 0.174/0.064	0	0
Yahoo! R3	DRUP	1, 5, 20, 100	0.065/0.065, 0.065/0.065, 0.066/0.064, 0.174/0.064	0	0
KuaiRec	IPS	5, 50, 200	0.001/0.001, 0.001/0.001, 0.133/0.001	1616	0
KuaiRec	DR	5, 50, 200	0.000/0.000, 0.000/0.000, 0.500/0.000	2819	0
KuaiRec	DRUP	5, 50, 200	0.000/0.000, 0.000/0.000, 0.500/0.000	2821	0

Table S11: Jointly private DRUP for all fixed denoising ranks ($\delta = 10^{-5}$).

Data	rank	$\epsilon = 0.5$	1	2	4	8	16	∞
Coat	8	0.381	0.385	0.400	0.399	0.398	0.449	0.533
	32	0.382	0.385	0.403	0.385	0.401	0.438	0.548
	128	0.372	0.370	0.387	0.401	0.389	0.422	0.536
	full	0.376	0.389	0.378	0.393	0.394	0.417	0.532
	non-private DRUP (all nuisances) 0.555; popularity 0.527; 29 public users							
Yahoo! R3	8	0.483	0.492	0.491	0.493	0.492	0.493	0.492
	32	0.481	0.490	0.491	0.493	0.492	0.493	0.492
	128	0.479	0.490	0.490	0.494	0.493	0.492	0.493
	full	0.473	0.488	0.491	0.494	0.494	0.493	0.492
	non-private DRUP (all nuisances) 0.652; popularity 0.525; 1540 public users							
KuaiRec	8	0.175	0.179	0.173	0.427	0.603	0.618	0.634
	32	0.174	0.175	0.154	0.412	0.598	0.618	0.634
	128	0.160	0.173	0.172	0.365	0.581	0.614	0.634
	full	0.182	0.202	0.223	0.280	0.377	0.492	0.634
	non-private DRUP (all nuisances) 0.637; popularity 0.577; 717 public users							

Table S12: Semi-synthetic KuaiRec with propensities estimated by the logistic exposure model (columns as in Table 4; regret not computed).

p	Nuisances	C	Estimator	RMSE T	utility error (%)		regret	nDCG@20
					random	popular		
est.	independent log	fixed	IPS	77.7	+361.4±5.6	+138.2±1.1	–	0.057
			IPS+WC	12.2	+278.9±2.0	+107.8±0.9	–	0.057
			DR	92.2	+370.0±7.6	+146.3±1.1	–	0.029
			DRUP	21.2	+277.0±2.6	+107.2±0.9	–	0.038
est.	cross-fitted	fixed	IPS	78.4	+359.9±6.9	+136.1±1.0	–	0.056
			IPS+WC	12.0	+274.7±2.1	+105.8±0.7	–	0.056
			DR	113.8	+430.5±8.2	+155.3±1.1	–	0.021
			DRUP	27.2	+307.1±2.5	+114.9±0.8	–	0.030
est.	sample split	fixed	IPS	89.5	+536.5±6.4	+315.1±2.0	–	0.066
			IPS+WC	12.0	+453.5±2.6	+280.8±1.8	–	0.071
			DR	152.9	+544.2±8.7	+322.3±2.4	–	0.020
			DRUP	23.4	+446.6±3.0	+276.0±2.0	–	0.029
Five hops (independent log, true p , fixed C): relative bias of T_5 , DR 3.46, DRUP 0.32.								

Table S13: Semi-synthetic KuaiRec with a sparser log (density 3%, 58% of the propensities below the clip $\tau = 0.01$), true and estimated propensities (columns as in Table 4; regret not computed).

p	Nuisances	C	Estimator	RMSE T	utility error (%)		regret	nDCG@20
					random	popular		
true	independent log	fixed	IPS	326.6	+49.1±26.9	+23.2±1.1	0.816	0.083
			IPS+WC	11.4	-27.9±2.5	-12.1±0.5	0.816	0.083
			DR	341.1	+331.4±28.5	+121.2±1.7	0.839	0.037
			DRUP	39.9	+163.6±5.4	+64.2±1.2	0.841	0.046
true	independent log	$\hat{Y}$	IPS	—	-49.3±2.3	-42.4±0.4	—	0.086
			IPS+WC	—	-64.7±1.2	-54.2±0.2	—	0.086
			DR	—	-8.9±2.5	-30.7±0.4	—	0.159
			DRUP	—	-30.2±1.3	-42.9±0.3	—	0.144
true	cross-fitted	fixed	IPS	326.6	+49.1±26.9	+23.2±1.1	0.816	0.083
			IPS+WC	11.4	-27.9±2.5	-12.1±0.5	0.816	0.083
			DR	521.4	+484.6±30.7	+166.3±3.5	0.851	0.030
			DRUP	73.3	+260.3±8.1	+108.2±3.0	0.848	0.040
true	cross-fitted	$\hat{Y}$	IPS	—	-75.2±0.8	-57.9±0.1	—	0.098
			IPS+WC	—	-78.3±0.7	-61.4±0.1	—	0.098
			DR	—	-43.4±1.2	-46.6±0.1	—	0.140
			DRUP	—	-51.4±0.9	-52.1±0.1	—	0.133
true	sample split	fixed	IPS	291.2	+200.4±26.7	+197.0±5.6	0.855	0.052
			IPS+WC	9.0	+123.3±3.9	+158.9±5.3	0.855	0.049
			DR	3058.9	+1572.9±142.3	+848.2±69.1	0.865	0.020
			DRUP	294.4	+932.8±38.9	+583.1±29.9	0.863	0.029
true	sample split	$\hat{Y}$	IPS	—	-10.1±5.3	-19.5±1.4	—	0.156
			IPS+WC	—	-43.8±1.6	-39.6±0.7	—	0.141
			DR	—	+24.8±5.1	-8.5±1.4	—	0.121
			DRUP	—	-13.3±1.6	-28.7±0.8	—	0.132
est.	independent log	fixed	IPS	930.7	+333.6±32.8	+122.0±1.7	—	0.073
			IPS+WC	39.3	+121.1±5.0	+51.8±1.0	—	0.073
			DR	957.3	+891.6±39.0	+268.3±2.3	—	0.033
			DRUP	95.0	+516.4±10.6	+171.7±1.8	—	0.040
est.	cross-fitted	fixed	IPS	798.1	+331.0±33.1	+119.1±1.7	—	0.073
			IPS+WC	37.4	+112.7±4.7	+48.4±0.9	—	0.073
			DR	1074.9	+1242.2±48.4	+322.0±4.6	—	0.026
			DRUP	148.0	+703.3±18.2	+222.2±4.1	—	0.036
est.	sample split	fixed	IPS	940.7	+753.3±31.7	+514.8±10.0	—	0.062
			IPS+WC	40.0	+502.0±8.2	+415.5±8.9	—	0.064
			DR	14661.2	+2972.1±290.9	+1342.7±83.3	—	0.019
			DRUP	1239.1	+1836.6±68.0	+964.5±35.5	—	0.028

Table S14: AUC of the candidate list on unbiased test data (mean $\pm$ std over splits; ties count one half). Same methods, tuning and splits as Table 5, which tuned on the metric named there.

Method	Coat AUC	Yahoo! R3 AUC	KuaiRand AUC	KuaiRec AUC
<i>Non-personalised</i>				
Popularity	0.644	0.655	0.581	0.765
Imputation only	0.654	0.652	0.546	0.800
<i>Trained, pointwise loss</i>				
MF	0.575	0.640	0.528	0.802
IPS-MF	0.570	0.643	–	0.789
DR-MF	0.620	0.645	0.549	0.801
DR-JL	0.584	0.656	0.549	0.797
MRDR	0.589	0.657	–	0.793
iALS	0.667	0.761	0.596	0.727
LightGCN	0.632	0.705	0.559	0.799
DR-LightGCN	0.658	0.675	–	0.781
<i>Trained, pairwise or sampled loss</i>				
MF (BPR)	0.614	0.751	0.603	0.782
PDA	0.612	0.758	–	0.781
MACR	0.639	0.729	0.575	0.762
LightGCN (BPR)	0.657	<u>0.771</u>	0.614	0.785
r-AdjNorm	0.658	0.775	0.617	0.778
NAVIP	0.653	0.771	<u>0.614</u>	0.796
SimGCL	0.646	0.736	0.578	0.765
<i>Training-free, logged graph</i>				
linear LightGCN / r-AdjNorm	0.648	0.759	0.599	0.783
EASE	0.614	0.745	0.569	0.623
GF-CF	0.638	0.756	0.603	0.789
BSPM	0.665	0.756	0.604	0.789
<i>Training-free, debiased graph</i>				
IPS adjacency (NAVIP-style)	0.641	0.756	0.600	0.786
IPS + walk correction	0.641	0.756	0.600	0.786
propagation over $\hat{Y}$ only (no residual)	0.656	0.663	–	0.807
EASE on DR graph	0.663	0.702	0.545	0.788
GF-CF on DR graph	0.664	0.739	0.595	0.808
BSPM on DR graph	<u>0.666</u>	0.737	0.601	0.808
DR adjacency	0.652	0.756	0.600	<u>0.809</u>
DRUP	0.653	0.756	0.600	0.809
DR adjacency, 5 hops	0.655	–	–	0.808
DRUP, 5 hops	0.649	–	–	0.809
<i>Training-free, sample split (Assumption 2 holds)</i>				
DR adjacency, sample split	0.651	0.710	0.592	0.806
DRUP-split	0.650	0.710	0.592	0.806

Table S15: Recall at the cut-off of Table 5 (@5 on Coat and Yahoo! R3, @10 on KuaiRand, @20 on KuaiRec). On Yahoo! R3 the lists have ten candidates, so Recall@10 is 1 for every method and Recall@5 is reported.

Method	Coat Recall@5	Yahoo! R3 Recall@5	KuaiRand Recall@10	KuaiRec Recall@20
<i>Non-personalised</i>				
Popularity	0.644	0.679	0.509	0.166
Imputation only	0.663	0.678	0.480	0.186
<i>Trained, pointwise loss</i>				
MF	0.579	0.670	0.469	0.189
IPS-MF	0.573	0.671	–	0.192
DR-MF	0.624	0.675	0.481	0.188
DR-JL	0.593	0.688	0.481	0.194
MRDR	0.592	0.689	–	0.193
iALS	0.681	0.790	0.531	0.183
LightGCN	0.638	0.731	0.494	0.193
DR-LightGCN	0.671	0.704	–	0.190
<i>Trained, pairwise or sampled loss</i>				
MF (BPR)	0.622	0.778	0.531	0.178
PDA	0.623	0.787	–	0.176
MACR	0.645	0.750	0.513	0.179
LightGCN (BPR)	0.659	0.799	<u>0.542</u>	0.180
r-AdjNorm	0.662	0.802	0.543	0.180
NAVIP	0.661	<u>0.800</u>	0.541	0.185
SimGCL	0.656	0.755	0.508	0.162
<i>Training-free, logged graph</i>				
linear LightGCN / r-AdjNorm	0.646	0.781	0.529	0.181
EASE	0.620	0.767	0.509	0.154
GF-CF	0.657	0.777	0.532	0.183
BSPM	0.675	0.779	0.531	0.183
<i>Training-free, debiased graph</i>				
IPS adjacency (NAVIP-style)	0.645	0.783	0.528	0.185
IPS + walk correction	0.645	0.783	0.528	0.185
propagation over $\hat{Y}$ only (no residual)	0.664	0.699	–	0.192
EASE on DR graph	0.668	0.720	0.487	0.180
GF-CF on DR graph	0.672	0.760	0.526	0.194
BSPM on DR graph	<u>0.677</u>	0.756	0.528	0.194
DR adjacency	0.654	0.783	0.528	0.194
DRUP	0.655	0.783	0.528	<u>0.194</u>
DR adjacency, 5 hops	0.662	–	–	0.194
DRUP, 5 hops	0.655	–	–	0.194
<i>Training-free, sample split (Assumption 2 holds)</i>				
DR adjacency, sample split	0.659	0.745	0.521	0.193
DRUP-split	0.658	0.744	0.521	0.193

Table S16: Which nuisance breaks the correction under cross-fitting. Relative absolute bias of the three-hop term (40 users, 60 items, true propensities, 1,500 replications) with the degree weights C fixed a priori (degrees of Y), taken from the imputation $\hat{Y}$, or from the edge estimates W ; nuisances from an independent log, cross-fitted, fitted in sample, or from the 20% sample split. Compare IPS and DR with and without the walk correction.

Nuisances	Degree weights C	IPS, uncorrected	IPS+WC	DR, uncorrected	DRUP
independent log	fixed a priori	1.10	0.05	0.81	0.04
	from $\hat{Y}$	2.20	0.13	1.95	0.11
	from W	5.31	0.33	3.58	0.22
cross-fitted	fixed a priori	1.10	0.05	0.89	0.05
	from $\hat{Y}$	0.39	0.64	0.61	0.80
	from W	3.22	0.82	3.95	0.74
in-sample	fixed a priori	1.10	0.05	0.51	0.23
	from $\hat{Y}$	0.43	0.71	0.59	0.70
	from W	0.94	0.97	0.80	0.87
sample split 20%	fixed a priori	0.89	0.07	0.58	0.07
	from $\hat{Y}$	1.10	0.21	0.92	0.21

Table S17: Key comparisons under three tests. Δ : mean per-user difference of the dataset metric (first method minus second). p_{Holm} : user-level paired t -test, Holm-adjusted over all comparisons of the dataset. p_{NB} : corrected resampled t -test over the split means (Nadeau and Bengio, 2003), Holm-adjusted over the same family. p_{TOST} : two one-sided user-level tests of equivalence within ± 0.005 nDCG, with the margins ± 0.002 and ± 0.01 in brackets as sensitivity; a small value means that the two methods are equivalent. “Identical rankings”: the two operators select the same configuration on every split, so no test applies. The margin is a convention, not derived from the data. Sample-split rows use the first draw.

Comparison (first minus second)	Coat	Yahoo! R3	KuaiRand	KuaiRec
DRUP vs DR adjacency	+0.0014; 1.000 / 1.000 / 0.034 [0.381, $< 10^{-3}$]	identical rankings	identical rankings	-0.0001; 1.000 / 1.000 / $< 10^{-3}$ [$< 10^{-3}$, $< 10^{-3}$]
DRUP vs DR adjacency, 5 hops	-0.0081; 1.000 / 1.000 / 0.809 [0.956, 0.299]	—	—	-0.0003; 0.170 / 1.000 / $< 10^{-3}$ [$< 10^{-3}$, $< 10^{-3}$]
DRUP-split vs DR-split	-0.0033; 1.000 / 1.000 / 0.295 [0.651, 0.019]	-0.0012; 1.000 / 1.000 / 0.001 [0.271, $< 10^{-3}$]	+0.0002; 1.000 / 1.000 / $< 10^{-3}$ [$< 10^{-3}$, $< 10^{-3}$]	-0.0001; 1.000 / 1.000 / $< 10^{-3}$ [$< 10^{-3}$, $< 10^{-3}$]
DRUP vs GF-CF on DR graph	-0.0143; 0.106 / 1.000 / 0.984 [0.998, 0.842]	+0.0247; $< 10^{-3}$ / 1.000 / 1.000 [1.000, 1.000]	+0.0025; 0.043 / 0.317 / $< 10^{-3}$ [0.716, $< 10^{-3}$]	+0.0004; 1.000 / 1.000 / $< 10^{-3}$ [$< 10^{-3}$, $< 10^{-3}$]
DRUP-split vs DRUP	+0.0087; 1.000 / 1.000 / 0.625 [0.717, 0.456]	-0.0659; $< 10^{-3}$ / 0.045 / 1.000 [1.000, 1.000]	-0.0092; $< 10^{-3}$ / 0.068 / 0.996 [1.000, 0.315]	-0.0035; 0.004 / 1.000 / 0.051 [0.949, $< 10^{-3}$]
DRUP vs DR-JL	+0.0658; $< 10^{-3}$ / 1.000 / 1.000 [1.000, 1.000]	+0.1350; $< 10^{-3}$ / 0.049 / 1.000 [1.000, 1.000]	+0.0739; $< 10^{-3}$ / $< 10^{-3}$ / 1.000 [1.000, 1.000]	+0.0044; 0.003 / 1.000 / 0.292 [0.983, $< 10^{-3}$]
DRUP vs MRDR	+0.0599; 0.003 / 1.000 / 1.000 [1.000, 1.000]	+0.1323; $< 10^{-3}$ / 0.024 / 1.000 [1.000, 1.000]	—	+0.0046; 0.003 / 0.339 / 0.367 [0.987, $< 10^{-3}$]
DRUP vs LightGCN (pointwise)	+0.0073; 1.000 / 1.000 / 0.585 [0.691, 0.398]	+0.0616; $< 10^{-3}$ / 0.441 / 1.000 [1.000, 1.000]	+0.0443; $< 10^{-3}$ / 0.003 / 1.000 [1.000, 1.000]	+0.0103; $< 10^{-3}$ / 0.030 / 1.000 [1.000, 0.567]
DRUP vs iALS	-0.0372; 1.000 / 1.000 / 0.968 [0.978, 0.941]	-0.0065; 1.000 / 1.000 / 0.643 [0.867, 0.190]	-0.0031; 1.000 / 1.000 / 0.119 [0.744, $< 10^{-3}$]	+0.0249; $< 10^{-3}$ / 0.069 / 1.000 [1.000, 1.000]
DRUP vs r-AdjNorm	-0.0290; 1.000 / 1.000 / 0.920 [0.943, 0.868]	-0.0187; $< 10^{-3}$ / 1.000 / 0.999 [1.000, 0.981]	-0.0101; $< 10^{-3}$ / 0.485 / 0.999 [1.000, 0.526]	+0.0296; $< 10^{-3}$ / 0.040 / 1.000 [1.000, 1.000]